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| SENSITIVE |

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SUPERSEDING
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MILITARY SPECIFICATION

WASHING AND SCREENING PLANT, AGGREGATE AND SAND, ELECTRIC-MOTOR-DRIVEN, TRAILER MOUNTED, 150 TONS PER HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a trailer mounted, electric-motor-driven, aggregate washing and screening plant with an output capacity of 150 tons per hour.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3820

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SPECIFICATIONS

FEDERAL

O-E-760 - Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents.
QQ-S-781 - Strapping, Steel and Seals.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-V-119 - Varnish, Spar, Phenolic-Resin.
MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-1055 - Barrier Material, Waterproof, Flexible.
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories For Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal.
MIL-V-13811 - Varnish, Waterproofing, Electrical Ignition.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-19826 - Generator Sets, Diesel Engine, Alternating Current, For Facilities Construction: 10 KW through 1000 KW (Not Stocked).
MIL-C-21567 - Compound, Silicone, Soft Film.
MIL-T-22085 - Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-B-22191 - Barrier Material, Transparent, Flexible, Heat Sealable.
MIL-G-28554 - Generator Sets, Mobile Electric Power and Supplement Equipment: Packaging of.
MIL-B-46176 - Brake Fluid, Silicone, Automotive, All Weather Operational and Preservative, Metric.
MIL-P-53030 - Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186	- Cushioning Anchoring, Bracing, Blocking, and Waterproofing With Appropriate Test Methods.
MS51336	- Lunette and Coupler, Drawbar, Ring.
MS75021	- Connector, Receptacle, Electrical -12 Contact, Intervehicular, 28 Volt, Waterproof.

(Unless otherwise indicated, copies of Federal and military specifications and standards are available from the Naval Publications and Forms Center (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standards and Regulations.
Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 242	- Specification for High Strength Low Alloy Structural Steel (DOD adopted).
ASTM D 3951	- Commercial Packaging (DOD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Controls and Systems (DOD adopted).
NEMA ICS 2 - Industrial Control Devices, Controllers and Assemblies (DOD adopted).
NEMA ICS 6 - Enclosures for Industrial Controls and Systems (DOD adopted).
NEMA MG 1 - Motors and Generators (DOD adopted).
NEMA WC 3 - Rubber Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy.
NEMA WC 7 - Cross-Linked-Thermosetting Polyethylene-Insulated Wire and Cable for the Transmission and Distribution of Electrical Energy (DOD adopted).

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code (DOD adopted).

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings (DOD adopted).
SAE J551 - Performance Levels and Methods of Measurement of Electromagnetic Radiation From Vehicles and Devices (30 - 1000 MHz) (DOD adopted).
SAE J560 - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
SAE J1292 - Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring (DOD adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The washing and screening plant, aggregate and sand, shall consist of a pneumatic-tired trailer on which is mounted an electric-motor-driven scrubber, triple-deck vibrating screen, and sand dehydrator with a portable batching conveyor. All chutes on the washing and screening plant, piping and accessories required shall be supplied. Water, electricity, and aggregates shall be supplied from an outside source. When specified (see 6.2), the manufacturer shall provide a skid mounted diesel engine drive generator (see 3.9.8.4).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.2.1 and 6.3).

3.3 Standard commercial product. The washing and screening plant shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are a part of the manufacturer's standard commercial product, shall be included in the washing and screening plant being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.6. Safety. All rotating or reciprocating parts and all parts subject to high operational temperature, that are of such nature or are so located as to be or become a hazard to the operating or attending personnel, shall be substantially guarded or insulated to the extent necessary to eliminate the hazard.

3.7 Maintainability. The washing and screening plant design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance service in the field, which will be accomplished with the use of conventional general purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the washing and screening plant.

3.8 Transportability. The washing and screening plant, without major disassembly, shall be capable of being towed in and out of a landing ship tank (LST) class vessel for loading and unloading as shown in figure 1. The height of any component extended beyond LST loading clearance requirements shall be either removable or retractable to facilitate clear passage.

3.9 Design and construction. All portions of hoppers, chutes, plates, and metal surfaces subject to abrasive wear caused by aggregate movement shall be lined or faced with replaceable, abrasive-resistant, steel plates conforming to ASTM A 242 or better. All bolt heads subjected to the flow of material shall not extend beyond surfaces exposed to the material.

3.9.1 Revolving scrubber. The revolving scrubber shall consist of a structural steel frame on which is mounted a receiving hopper, scrubber drum, water system, and discharging chute. The scrubber shall be of sufficient size for the capacity of the plant specified in 3.10.1.

3.9.1.1 Receiving hopper. A receiving hopper shall be provided to receive material from a portable conveyor and feed the material into the scrubber drum by means of a chute. The receiving lip of the hopper shall be not more than 25 feet (ft) above the ground.

3.9.1.2 Scrubber drum. The scrubber drum shall be constructed of steel plates not less than 0.375 inch (in.) in thickness, and lined with steel liner plates not less than 0.25 inch in thickness. The drum shall be mounted on roller bearings and shall be set at a slope with the horizontal. The drum shall be kept or guided in position by means of trunnion roller or rollers with collar. The drum shall be equipped with not less than three removable steel baffles or vanes not less than 0.25 inch thick, arranged to propel the material away from receiving end and deliver it toward the discharge chute. The scrubber drum shall be rotated by an electric motor through roller chain, gears, or through friction trunnion rollers.

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3.9.1.3 Water system. A water inlet header equipped with a standard 6 inch, 150 pound (lb) flange, and a pressure regulating valve, shall be mounted on the plant to receive water from a connecting 6 inch steel pipe supply line. Pressurized water streams shall be evenly distributed to spray pipes and nozzles (see 3.9.5), suitably located in revolving scrubber drum and between screens, to wash out all the fine materials.

3.9.1.4 Discharging chute. A discharging chute shall be provided for discharging materials from the drum into the hopper of the vibrating screen. The material shall be discharged evenly over the width of the vibrating screen.

3.9.2 Vibrating screen. The vibrating screen shall be the inclined or horizontal type with three decks. Screen size shall be not less than 4 feet in width by 10 feet in length. The top deck shall have a one inch square opening screen cloth, the second deck 0.50 inch square opening, and the bottom deck 0.25 inch square opening for normal operation, but shall be capable of being converted to allow installing other screen opening sizes for meeting different job requirements. The vibrating screen shall consist of a stationary frame upon which shall be mounted a live frame with screen cloths, a vibrating mechanism, a screen feed hopper, and a screen discharge hopper and chutes.

3.9.2.1 Stationary frame. The stationary frame of the screen shall be constructed of steel plates on structural steel members of rigid construction with sufficient crossmembers to securely hold proper width to insure proper fittings of screen cloths. The live frame shall provide means for holding the replaceable screen cloth securely in place to prevent movement of the cloth during the full-load operation. A series of longitudinal support bars shall be provided which shall create a crown of not less than 0.50 inch in the screen cloths. A molded rubber cushion shall be provided between the screen cloths. The molded rubber cushion shall be capable of being removed and replaced by the use of standard tools. The frame shall be balanced to provide an even flow of material across the full width of the cloth. The live frame shall be mounted on the stationary frame by means of springs, supporting rubbers, or by other means producing the equivalent in performance.

3.9.2.2 Screen cloth. All screen cloth shall be made of drawn and hardened, oil tempered wire with a carbon range between 0.55 percent (%) to 0.70 percent. Two complete sets of screen cloth of the sizes specified in table I, in two or more sections and of equal length, shall be furnished with each washing and screening plant.

3.9.3 Vibrating mechanism. A mechanical vibrating mechanism shall be provided for activating the live frame assembly. The mechanism shall be equipped with a replaceable protective cover to prevent entry of abrasive material. The mechanism shall be driven by an electric motor.

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TABLE I. Wire size.

Minimum clear opening inches	Wire diameter inches
2.50	0.50
2.00	0.4375
1.50	0.375
1.00	0.3125
0.75	0.25
0.50	0.192
0.375	0.162
0.25	0.135
0.0787	0.035

3.9.3.1 Bearings. All shaft bearings on the vibrating mechanism shall be double row, self-alining, roller bearings. All bearings exposed to dust shall be provided with labyrinth type or equivalent, grease, oil, and dust seals to prevent leakage of lubricant and to prevent dust from entering the bearings. Drive shaft bearings shall withstand the loads imposed under maximum operating conditions.

3.9.3.2 Feed hopper. A hopper shall be mounted over the vibrating screen to receive material from the scrubber chute and discharge the material evenly across the full width of the vibrating screen.

3.9.4 Screen discharge hopper and chutes.

3.9.4.1 Discharge hopper. A discharge hopper shall be located under the vibrating screen to receive all material from the bottom deck and pass it into the receiving hopper of the conveyor.

3.9.4.2 Chutes. Material chutes shall be provided for the three screen decks, and the ground clearance under the chutes or stub conveyors, if furnished, shall be not less than 42 inches to permit operating clearance for the concentrating hoppers or portable conveyors. The chute arrangement shall provide for not less than 4.5 feet clearance between centerlines of discharging end of the chutes.

3.9.4.3 Overside discharge chute. An overside discharge chute shall be provided on the downstream end of the top screen for rejecting the larger aggregates after washing on the top screen.

3.9.5 Spray pipes. The vibrating screen shall be equipped with spray pipes above each screen deck. The spray pipes shall contain a multiple of spray nozzles directed downward. The spray pipes shall receive water under pressure from the main water header controlled by a valve, and shall be provided with valves for control of water volume on each deck of the screen.

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3.9.6 Washer-classifier-dehydrator. The washing and screening plant shall be provided with a washer-classifier-dehydrator capable of passing not less than 35 percent of the fine materials through the fine screen. The washer-classifier-dehydrator shall consist of a receiving hopper, including settling area with an adjustable water overflow weir, a single or twin screw, or paddle type conveyor, and a sand receiving hopper and discharge chute.

3.9.6.1 Receiving hopper. The receiving hopper shall be of sufficient size and shape to receive the fines passing the screen. The bottom of the settling area and trough shall be shaped to the conveyor and provide minimum operating clearance to prevent the material from backsliding. A water supply shall be provided to flush the drain trough. An adjustable overflow weir shall be provided to regulate the depth of water in the settling area. A quick-opening valve shall be provided to entirely drain the settling area.

3.9.6.1.1 Conveyor. The receiving hopper shall be provided with a conveyor system to transfer the settled sand from the settling area to the discharge chute. The conveyor system shall gradually elevate the sand from the water up the inclined trough, designed to fit the conveyor, in order to drain off excess water before depositing the sand into the chute. The washer-classifier-dehydrator shall be powered by means of an electric motor. All bearings shall be of the roller or sleeve type, completely enclosed in an oiltight case, with positive means to exclude water and abrasives. The conveyor system shall be of the single or twin screw auger type. The screw conveyor shall be provided with replaceable iron or steel cast segmental spiral flight sections having a Brinell hardness number of not less than 450. The screw conveyor and settling areas shall be set at an angle of between 15 degrees (°) and 20° and extend sufficiently above the settling area water level of drain excess water from the fines and discharge into the sand chute.

3.9.7 Trailer. The washing and screening plant shall be securely mounted on a two or three axle trailer. The gross weight on any one axle shall be not greater than 18,000 pounds (lbs). The frame shall be fabricated from structural steel, welded, bolted, riveted, or a combination thereof, with sufficient braces or reinforcements to provide a rigid support under all operating and travel conditions. The front axle shall have fifth-wheel type steering capable of being turned 90° right or left of the axis of travel. The drawbar shall be hinged and adequately braced to withstand all towing and turning forces and shall be provided with a lunette ring conforming to MS51336 and two safety chains. The trailer shall be equipped with straight air or air over hydraulic service brakes controlled from the towing vehicle cab. All necessary air hose and standard air couplings for connecting to the brake system of the towing vehicle shall be provided. The service brake installation shall include a breakaway system capable of automatically applying brakes if the trailer should accidentally be separated from the towing vehicle, and shall be capable of maintaining the application for not less than 15 minutes. Not less than four screw jacks shall be provided, two at each end of the trailer frame, located so that the jacks will support the entire unit while leveling and blocking the washing and screening plant. The screw jacks shall retract sufficiently to preclude interference with the unit

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while traveling. The washing and screening plant shall be capable of being towed on a paved surface at a speed of 20 miles per hour (mph) and over cross-country terrain at a speed of not less than 5 mph.

3.9.7.1 Lighting. Unless otherwise specified (see 6.2), the electrical lighting system shall be 12 volts (V) potential. The lighting system shall conform to DOT Motor Carrier Safety regulations. All lights and reflectors shall be protected from operational hazards by mounting in recessed or vertical surfaces of the rub rails or trailer bumpers. The front of the trailer shall be equipped with a receptacle conforming to SAE J560, with the receptacle located and the conductors connected and color coded as specified therein. All electrical wiring shall conform to SAE J1292.

3.9.7.1.1 Receptacle system, 24V. When specified (see 6.2), the trailer shall be equipped with one 7-contact receptacle with cover and one 12-contact receptacle with cover installed at the front of the trailer. The 7-contact receptacle shall conform to SAE J560 round socket for jumper cable. The 12-contact receptacle shall conform to MS75021, part number MS75021-1. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12V directly to the trailer circuits. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of a tactical military design towing vehicle from 24V to 12V. The 24V, 12-contact shall be connected through the necessary resistors into the 12V trailer circuits as specified in table II.

TABLE II. 12-contact receptacle connections to trailer circuits.

12-contact receptacle pin	Trailer circuits
A	Do not connect
B 1/	Connect to circuit number 3
C	Do not connect
D	Connect to circuit number 1
E	Connect to circuits numbers 2 and 6
F	Do not connect
H	Do not connect
J 1/	Connect to circuit number 5
K	Connect to circuit number 7
L	Do not connect
M	Do not connect
N	Do not connect

1/ Circuits B and J on tactical truck-tractors are combination stop and turn indicator circuits. The normal 12V turn signal lights will function both as turn signals and stoplights and the normal 12V stoplights will not be in operation when the trailer is connected to a towing vehicle with a 24V power supply. Because of this condition, the number 4 stoplight circuit is not connected to the 24V, 12-contact receptacle.

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3.9.7.2 Tires. Tires shall be of high quality commercial type. Tires shall have individual rated load-carrying capacities equal to the maximum individual tire loading imposed by the operation in accordance with the recommendations of the TRA Yearbook. Tires shall have nondirectional type tread.

3.9.8 Electric equipment.

3.9.8.1 Electric motors. All electric motors shall be polyphase, single speed, induction type, and connected for 460 volts alternating current (Vac), 3-phase, 60 Hertz (Hz) operation. All motors shall be totally enclosed, nonventilated, or totally enclosed fan-cooled machines conforming to NEMA MG 1. Motors shall be suitable for the respective applications in an ambient temperature of 122 degrees Fahrenheit (^{o}F).

3.9.8.2 Motor control center. The motor control center shall be a free standing, outdoor type, combination motor control unit rated for 480 Vac, 3-phase, 60 Hz, and shall conform to NEMA ICS 1, class II, type B, enclosed in a NEMA ICS 6, type 3S enclosure. Provision shall be made for remote control stations (start-stop pushbuttons) where required. The motor control center, complete with all motor controllers, components, and accessories, shall be suitable for continuous operation under conditions as follows:

a. At ambient temperatures between +32°F to +120°F and at any possible relative humidity.
b. Exposure to severe dust conditions.
c. Exposure to salt air.
d. At any altitude from mean sea level to 5,000 feet above mean sea level.

The motor control center shall have a three pole, manually operated main circuit breaker with overload and short circuit protection, on the incoming power supply to the control center. The termination for the incoming power supply to the main circuit breaker shall have screw pressure type terminals for not less than three sets of power supply cables. One set of terminals (one per phase) shall be suitable for terminating conductors from number 4 American Wire Gage (AWG) to number 1 AWG. The motor control center shall have an instruction plate (see 3.19.1) permanently affixed in a conspicuous location. In addition to instructions for connecting and operating the washing and screening plant, the instruction plate shall show the power supply requirement of the washing and screening plant in bold type, as follows:

POWER SUPPLY REQUIRED
460 VOLTS, 3-PHASE, 60 HERTZ
INPUT CAPACITY, MAXIMUM _____ KVA (_____ KW)
OPERATING LOAD (AT RATED CAPACITY) _____ KVA (_____ KW)

Electrical power may be supplied from either a commercial power source or diesel engine driven generator set conforming to MIL-G-19826 (see 3.9.8.4).

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3.9.8.2.1 Motor controllers. Motor controllers (starter) shall be alternating current, general purpose, Class A, magnetic controllers for single speed, nonreversing, nonjogging, squirrel-cage induction motors, conforming to NEMA ICS 2. Controllers for motors 15 horsepower (hp) and larger shall be reduced voltage or current motor controllers. Controllers for motors less than 15 hp may be full-voltage (across-the-line) controllers. For consistency in washing and screening plant operation, the controllers shall be interlocked to provide sequence starting, with the larger motors to be started first, in order to reduce the required power input capacity to a minimum. The starting sequence shall be included in the connecting and operating instructions.

3.9.8.3 Power and control cables. All power, control and grounding cables, fittings, and accessories required to completely connect the washing and screening plant for operation shall be furnished in the length required for each circuit. The power cable (one three-conductor cable or three one-conductor cables each), not less than 100 feet in length, shall be furnished to connect the motor control center to the incoming power supply source. All power and control cables, for the power supply and for interconnecting the washing and screening plant, shall be heavy-duty portable cables, conforming to NEMA WC 3 or NEMA WC 7 and suitable for direct burial or laying on the earth surface. All portable power cables shall be rated for conductor operating temperature of not less than 167°F. The load on any conductor shall not be greater than the allowable ampacities of NFPA 70 for direct burial conductors. The grounding system for the complete washing and screening plant shall include the grounding conductors, fittings, ground rods, and clamps. The grounding conductors shall be of suitable size and type for the application.

3.9.8.4 Skid mounted diesel engine driven generator set. When required (see 3.1), the manufacturer shall provide a type II, style A, class 1 skid mounted diesel engine driven generator set (referred hereinafter as the generator set) conforming to MIL-G-19826. The generator set shall be equipped with a weatherproof housing. The housing shall protect the generator set for both storage and operation in all weather conditions. The generator set shall be capable of providing all power requirements at the voltages specified in 3.9.8.1 to operate all electrical motors and accessories of the washing and screening plant under the loads specified in 3.10.

3.9.9 Portable stockpile conveyors. Four portable stockpile conveyors shall be provided and shall consist of a conveyor frame with a conveyor belt not less than 24 inches wide, and not less than 60 feet in length, measured from the lower pulley to the head pulley with masts or arms for elevating the conveyor frame to an angle up to 20°; a two wheel pneumatic-tire chassis; and an electric motor to provide motive power. The capacity of the motor shall be sufficient to handle the amount and load of crushed rocks produced and be so mounted that the power is transmitted directly to the head pulley. The conveyor belt drive shall be designed to prevent rollback of the belt in the event of power failure. The conveyor frame shall be raised either hydraulically or by power and shall be capable of being locked at various angles. The removable receiving hopper, having a capacity of not less than

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7 cubic feet, shall be installed on the lower end of the conveyor. The chassis shall be designed so that the entire unit may be towed over the road at a speed of not less than 20 mph. The chassis drawbar shall be equipped with a lunette ring conforming to MS51336 and two safety chains. The chassis wheels shall be capable of being pivoted not less than 90° in either direction and locked in any position.

3.9.10 Portable feed conveyor. One portable feed conveyor shall be provided and shall be constructed similar to, but not necessarily identical to, the portable stockpile conveyors specified in 3.9.9, except that the width of the conveyor belt shall be not less than 30 inches.

3.10 Performance.

3.10.1 Capacity. The washing and screening plant shall be capable of processing not less than 150 tons per hour of aggregate in the size range of 2.5 inches and smaller and containing not more than 35 percent fines passing a 0.25 inch screen during quarry or gravel pit operations.

3.10.2 Towing. Each wheeled component, when towed, shall be capable of travel over smooth hard-surfaced primary highways at speeds up to 20 mph, travel over unimproved roads and hilly cross-country terrain at speeds up to 5 mph, and negotiate a 90° turn without weaving.

3.10.2.1 Braking ability. The combined service brakes of the towing and towed vehicle shall control, decelerate, and stop the combination in conformance with DOT Motor Carrier Safety Regulations requirements. When actuated, the emergency breakaway system shall produce smooth emergency brake application.

3.11 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 lbs per square inch (psi) or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated as specified in section 5.

3.12 Electromagnetic interference suppression. When specified (see 6.2), the washing and screening plant and when provided (see 3.1), the generator set, shall meet the electromagnetic interference requirements and test limits for equipment as specified in SAE J551.

3.13 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points shall not be coated.

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3.14 Lifting and tiedown attachments. The washing and screening plant shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the washing and screening plant. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown markings shall show which attachments are intended for the tie down of the washing and screening plant on the carrier when shipped. Lifting devices shall be designed to exclude the use of spreader bars.

3.15 Toolbox. A closed compartment or toolbox shall be provided to carry those tools and accessories required to operate the washing and screening plant. Toolbox, when furnished, to hold tools and accessories shall be made of sheet steel having a nominal thickness of 0.0747 inch (US revised standard gage number 14). The toolbox shall have a hinged lid and a trunk drawbolt of a type that will keep the lid closed when the toolbox is subjected to vibration. The toolbox shall be securely fastened to the washing and screening plant in a protected and accessible location.

3.15.1 Special maintenance tools. Such special tools as are required for field maintenance and operation of the equipment shall be packed in the toolbox and shipped with the equipment. Special tools are defined as those not normally included in mechanic's tool kit.

3.16 Cleaning, treatment, and painting. Surfaces which are to be painted shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, greases, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to the manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.17 Data plates.

3.17.1 Instruction plates. The washing and screening plant shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 0.125 inch in diameter.

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3.17.2 Nameplate. A nameplate will be furnished by the contracting officer for each component unit. The contractor shall stamp all necessary data in the blank spaces provided for that purpose, and securely affix it to each washing and screening plant in a conspicuous place with nonferrous screws, bolts or rivets not less than 0.125 inch in diameter. The applicable nomenclature contained in contract item description shall be placed in the top blank.

3.18 Servicing and adjusting. Prior to acceptance of the washing and screening plant by the Government, the contractor shall service and adjust the washing and screening plant for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alinement.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

3.19 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.20 Workmanship.

3.20.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the metal, may be used instead of shearing and sawing. All bends shall be made by controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating and heated metal shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.20.2 Bolted connections. Boltholes shall be punched or drilled within manufacturer's design tolerances and shall have the burrs removed. Washers or lockwashers shall be provided, and all bolts, nuts, and screws shall be tight.

3.20.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

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3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.20.5 Castings. All castings shall be sound and free from patching misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one washing and screening plant when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1 through 4.4.6, and the packaging inspection of 4.5. The first article may be a standard production item from the contractor's current

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inventory provided the washing and screening plant meets the requirements of this specification and is representative of the design, construction, and manufacturing techniques applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. Each washing and screening plant produced shall receive the examination of 4.3, the test of 4.4.3, and the packaging inspection of 4.5.

4.3 Examination. Each washing and screening plant shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The component units shall be arranged in the order as shown in the approved arrangement plan of the washing and screening plant and the piping and accessories shall be connected to the component units for compatibility and subjected to the tests in accordance with 4.4.1 through 4.4.6. Nonconformance to the requirements specified in section 3 shall constitute failure and cause for rejection.

4.4.1 Test conditions. Prior to being tested, the washing and screening plant shall be lubricated. Oils shall be those designated for use in the ambient temperature at which the tests are being performed. During all washing and screening plant tests, the motors shall be run at normal operating speeds. Where individual components are tested with the use of power sources other than the washing and screening plant power supply, the component will be operated at the normal washing and screening plant operating speeds. As services and adjustments are performed, the washing and screening plant shall be checked for ease of maintenance.

4.4.2 Compatibility. Assemble and operate all the component units in the arrangement as shown in the approved arrangement plan. The test may be conducted with load or without loads for not less than 0.5 hour each, provided that operating with or without loads will cause no damage to any component unit or to the washing and screening plant. All component units shall assemble and operate compatibly without modifications. The piping, hose, valves, fittings, and accessories shall be assembled, and matchmarks shall be painted on each of the piping, fittings, valves, and accessories, before disassembled for shipment, for the purpose of easy identification when reassembled in the field.

4.4.3 Cleaning ability and capacity test. The washing and screening plant shall be tested for the capability of washing and screening not less than 150 tons per hour of a sufficient quantity of 2.5 inch, 1.5 inch, 0.75 inch, 0.50 inch, 0.375 inch, and 0.25 inch crushed rock or bank run gravel containing not more than 35 percent of fine materials.

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4.4.4 Towing test. The washing and screening plant and the portable conveyor shall be towed by a wheeled tractor or truck tractor at a speed not greater than 20 mph, over smooth hard-surfaced roads for a distance of not less 5.0 miles, then towed cross-country for not less than 0.5 mile. During both runs, not less than five right-angle turns in both directions, and not less than 10 brake applications shall be made.

4.4.5 Lifting and tying down attachment tests. When lifting and tying down attachments are required, the washing and screening plant shall be tested to conform to 3.14.

4.5 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts and assemblies vulnerable to pilferage, damage, and loss; to accomplish reduction in cube; and to meet carrier limitations of height, width, and weight. Removed nuts, screws, pins, and washers shall be installed in mating parts and secured to prevent loss. Gaskets and related items shall be individually preserved Method IC or III as required and placed in the toolbox or packaged with other removable parts. Keys shall be secured in keyways of the primary components, attached with shipping documentation, or packaged separately. Disassembly should be limited to parts and components easily removed and installed using no special tools or skilled personnel. Disassembly required to preserve equipment components shall be the minimum necessary to perform preservation.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.5.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.5.2 Machined surfaces. Exposed machined surfaces, shall be coated with type P-6 or P-11 preservative and wrapped or covered as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

5.1.1.6 Gears.

5.1.1.6.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or with primer conforming to TT-P-664 or MIL-P-53030.

5.1.1.6.2 Enclosed gears. Enclosed gears shall be filled to the operating level with the approved lubricant required for operation. Gear mechanism shall be operated to insure coating of all interior surfaces with preservative. The gear housing shall be identified with a weatherproof tag to indicate, "This housing is filled to the operating level with lubricant required for operation. Do not drain until first required lubrication change." Markings shall be applied to the tags with a waterproof material. The tags shall be attached in a conspicuous location.

5.1.1.7 Drive chains.

5.1.1.7.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified in 5.1.1.6.2 for enclosed gears.

5.1.1.7.2 Exposed chains. Exposed chains shall be coated with enough type P-9 preservative to ensure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the entire chain and unpainted surfaces of the sprocket shall be coated with type P-1 preservative.

5.1.1.8 Drive belts and pulleys. Drive belts and pulleys shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-C-53030. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.9 Hydraulic systems (except hydraulic brakes of steering systems). The hydraulic fluid supply tank shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be

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coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches into the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.10 Brakes.

5.1.1.10.1 General. Clean all surfaces as required to remove contaminants. Coat unpainted exterior surfaces of rods, levers, cables, clevises, linkage, and associated parts with type P-1 preservative. Remove brake drum covers and spray enclosed metallic components with primer TT-P-664 or MIL-P-53030. Spray coat brake drum friction surfaces with a very thin coat (0.5 to 0.6 mil thick) of primer. Coat friction surfaces of cams, adjusting screws, and anchor pins with type P-6 or P-7 preservative. Back off all brake shoe adjusters for maximum clearance between brake drum and brake shoes. Do not apply coating of any type to contact surfaces of electric brakes. Adjust brake shoes to provide maximum clearance to drum surfaces and lined surfaces. In a conspicuous location in the operator's compartment, attach a weatherproof warning tag marked, "Brakes preserved. Do not apply brakes when vehicle is being moved. Use tow bar or similar arrangement for moving vehicle. Adjust brakes before placing equipment into operation." Markings shall be applied with waterproof material.

5.1.1.10.2 Hydraulic brakes. Remove and disassemble the brake cylinder. Clean the cylinders, pistons, springs, and other metallic surfaces normally in contact with hydraulic brake fluid with alcohol conforming to O-E-760. Do not use solvents (e.g., C-3, C-7, or C-18). Flush hydraulic lines with alcohol. Follow this by cleaning process C-8. Eliminate any residue of the finger print remover. Dry by procedure D-1 or D-4. Coat the cylinders, pistons, springs, rubber cups, rubber dust boots, and other internal metallic and rubber parts with silicone protector, MIL-C-21567. Fill the hydraulic system and the hydraulic fluid supply reservoirs with silicone brake fluid, MIL-B-46176. Fill the hydraulic system and the hydraulic fluid supply reservoirs with silicone brake fluid, MIL-B-46176. Bleed as necessary to ensure the system is completely filled with fluid. After filling pressure bleeder reservoir, allow fluid to stand for 3 to 5 minutes for air bubbles to escape before replacing cap. In order to prevent air entrainment, if vehicle is manually bled, do not depress brake pedal over one-half of travel.

5.1.1.10.3 Air-actuated brake systems. The air-actuated brake systems shall be drained of condensation. Interior surfaces of the air supply tanks shall be fogged with type P-10, grade 30 preservative. Excess preservative

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oil shall be drained. Threaded openings and the threads of drain plugs shall be coated with Type P-10. Air line filters shall be drained and closed. The drain plugs shall be reinstalled. Air line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting caps or plugs as applicable conforming to MIL-C-5501 or by sealing the ports with waterproof tape. Attach a waterproof tag to the valves and service lines indicating, "Remove (plugs, caps, or tape) from exhaust ports and valves before operating." Markings shall be applied with a waterproof material.

5.1.1.11 Electrical components. Openings in sockets, coupling plugs, and switches shall be sealed with waterproof tape.

5.1.1.11.1 Electrical motors. Electrical motors shall be preserved level A by the alternate requirements of MIL-E-16298.

5.1.1.11.2 Electrical modules. Circuitry modules, printed circuit boards, and panel boards that are not hermetically sealed should be removed, if possible, and packaged method II. If the complexity of the circuitry will not allow removal, seal area with a waterproof barrier conforming to MIL-B-81705 or MIL-B-22191, type III, or tape conforming to MIL-T-22085.

5.1.1.12 Water spraying systems. Interior surfaces of water spraying systems, including pumps, piping, valves, and spray nozzles, shall be coated with type P-3 or P-21 preservative. All openings for which a closure is not provided shall be sealed with caps, plugs, gasketed blind flanges, or with waterproof tape.

5.1.1.13 Conveyor belts. Metal facings shall be coated with varnish conforming to TT-V-119 or MIL-V-13811. Each belt shall be completely rolled with a tube or piece of round wood placed in each end fold of the belt to prevent sharp bends. Prior to rolling, belt surfaces shall be dusted with powdered talc to prevent surfaces from adhering together. When required, to hold belt in compact roll, strapping conforming to QQ-S-781, class 1, type I, finish B, shall be placed around the roll. Suitable protective cushioning shall be placed under the strapping, and the strapping not tensioned so tight as to damage the belting. Each rolled belt shall be wrapped with waterproofed barrier conforming to PPP-B-1055, class E-1 or E-2, with all laps, seams, and folds sealed with adhesive conforming to MMM-A-260.

5.1.1.14 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10.0 lbs psi above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to 66.6% of the specified operating pressure.

5.1.1.15 Wire rope. Wire rope shall be preserved method I by coating with type P-1 preservative. The preservative shall be applied to all surfaces of the wire rope. Excess preservative shall be allowed to dry (not less than 4 hours). The wire rope shall be bundled or coiled and wrapped with a barrier material conforming to MIL-B-121, type I, grade A, class 2 or grade C and secured.

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5.1.1.16 Openings. Openings in the equipment which will trap free falling water shall be sealed by any suitable means to prevent the entry of water.

5.1.1.17 Lubrication. Lubricate all grease fittings with type P-11 preservative. Wheel bearings should be lubricated with a compound specified by the manufacturer. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-9.

5.1.1.18 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.19 Consolidation. The dash compartment and tool boxes shall be utilized for the consolidation of loose components, tools, and publications. Tool boxes shall be secured to prevent pilferage. The remainder of the components that will not fit in the dash compartment or tool boxes shall be consolidated, along with disassembled components, in boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. The contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186. Boxes shall be placed in the cab or secured to the equipment with appropriate strapping. Arrangement and location on the equipment shall be such so as not to increase cubage or interfere with lifting or mobility of the equipment.

5.1.2 Commercial. The complete washing and screening plant, and when provided (see 3.1), the generator set shall be preserved and packaged in accordance with ASTM D 3951. The washing and screening plant, and when provided (see 3.1), the generator set (see 3.1) shall be lubricated for operational service as required in the operator's manual.

5.2 Packing. The packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. The generator set, when provided (see 3.1), shall be packed as specified in MIL-G-28554. Packing for the washing and screening plant shall be as specified in the following paragraphs.

5.2.1.1 Basic washing and screening plant. The basic washing and screening plant shall be shipped uncrated. Small disassembled components for each washing and screening plant shall be packed in close-fitting boxes conforming to PPP-B-621, class 2; or PPP-B-601, military overseas type. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes. Containers shall be positioned and secured to the washing and screening plant by bolting or with strapping in such a manner as not to interfere with towing or lifting of the washing and screening plant with slings. The revolving scrubber, if removed, and any other large disassembled component shall be packed on a skid base. The base shall be constructed in accordance with the requirements for skid type base of MIL-C-104, style A, except that floorboards and headers do not require notching. The scrubber and any other components shall be anchored, blocked, and braced in accordance with MIL-C-104.

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5.2.1.2 Portable conveyors. Portable conveyors shall be shipped knocked down to the most practicable extent. The structural components shall be nested, arranged, and secured with bolts, or steel strapping, in combination with suitable wood blocking or battens, as required to form nonshifting bundles. The strapping shall be 0.035 inch by 1.25 inches strapping conforming to QQ-S-781, class 1, type I, finish B. The strapping shall be stapled to any wood blocking or battens provided. Where it is not practicable to disassemble the equipment to the extent required for suitable bundling, the sections shall be secured to skid bases conforming to the skid-type base of MIL-C-104, style A crates, and shipped as individual assemblies. The electric motors, conveyor belts, bolts, nuts, washers, and any other small components requiring the protection of a container for safe convenient handling shall be packed in containers as specified for disassembled components. Where skid bases are used, consideration shall be given to consolidating and securing any bundled or boxed components on the skid bases.

5.2.2 Commercial. The complete washing and screening plant, and when provided (see 3.1), the generator set shall be prepared for shipment in accordance with ASTM D 3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The washing and screening plant is intended for use in washing and screening crushed aggregates from screening pit gravels, bank runs, or dry river beds. The washing and screening plant can be used as a single independent unit or in combination with a crushing and screening washing and screening plant.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a type II, style A, class 1 skid mounted diesel engine driven generator set conforming to MIL-G-19826 is required (see 3.1 and 3.9.8.4).
d. When a first article sample and inspection is required (see 3.2 and 4.2.1).
e. When the electrical lighting system is other than 12V (see 3.9.7.1).
f. When the trailer is to have one 7-contact receptacle and one 12-contact receptacle installed (see 3.9.7.1.1).
g. Means for lubrication, if other than as specified (see 3.11).

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h. When electromagnetic interference suppression is required (see 3.12).
i. When treatment for fungus resistance is required (see 3.13).
j. Color required for finish coat (see 3.16).
k. Level of preservation and level of packing (see 5.1 and 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
---	DI-M-4022	Commercial Manual	---
---	DI-E-5325	Engineering Drawing (Commercial)	---
---	UDI-A-23264	Certification Data/Report	---

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3 First article. When a first article inspection is required, the item should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete washing and screening plant. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Subject term (key word) listing.

Portable stockpile conveyors
Revolving scrubber
Vibrating screen

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

Review activity:
Army - AT

User Activity:
Army - CE

Preparing activity:
Navy - YD

(Project 3820-0190)

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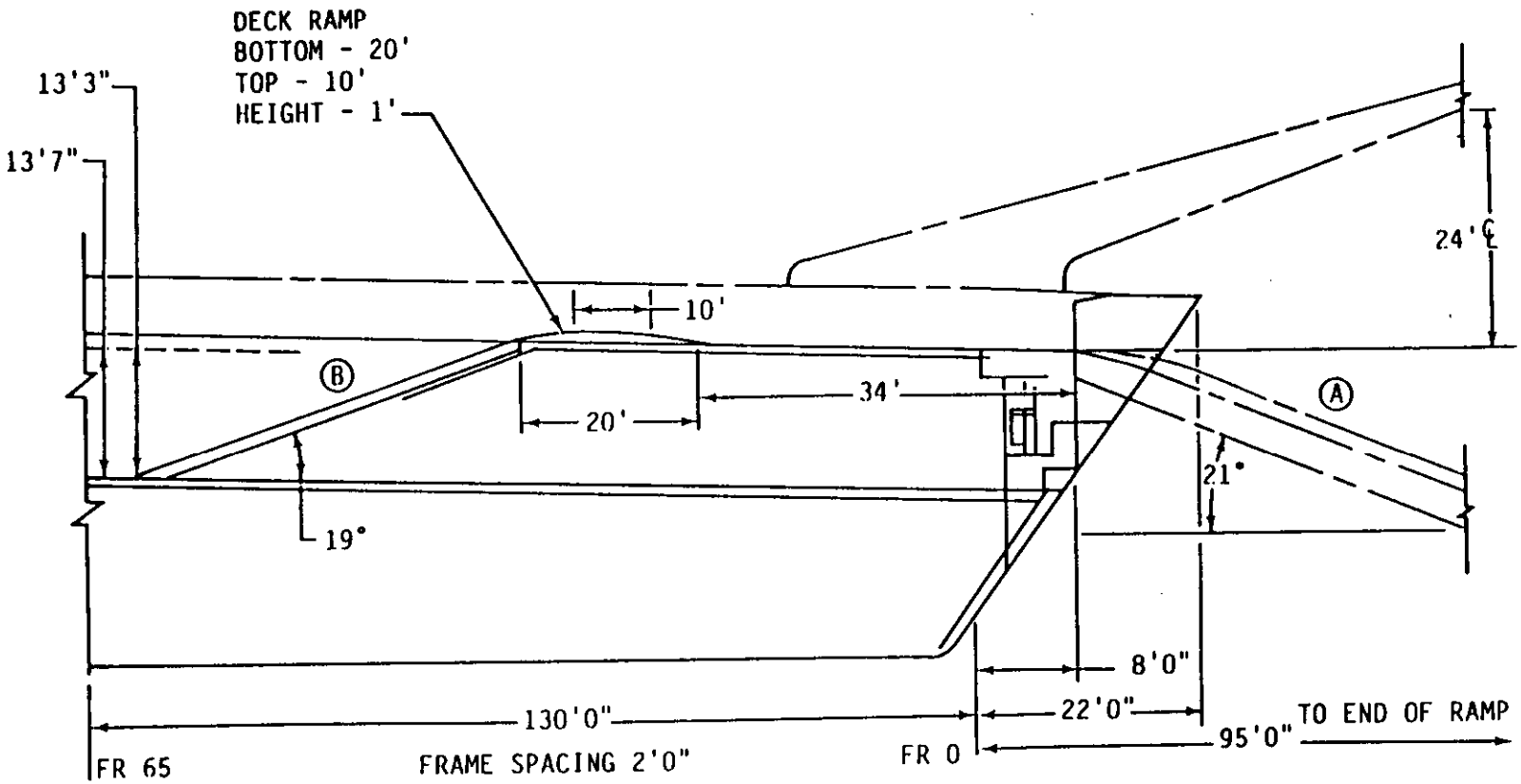


FIGURE 1. Profile, LST 1179 class bow.

LEGEND

BOW RAMP A
CAPACITY - 75 TON
LENGTH - 100'
WIDTH - 15"
MAX. RAMP ANGLE - 21°
CLEARANCE TO OVERHEAD - NOT
NOT APPLICABLE

BETWEEN DECK RAMP B
CAPACITY - 75 TON
CLEARANCE TO OVERHEAD:
RAMP DOWN - 13'3"
RAMP UP - 13'7"
WIDTH - 15'8"
LENGHT - 47'9"
RAMP ANGLE - 19°

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-23455D	MILITARY SPECIFICATION - WASHING AND SCREENING PLANT, AGGREGATE AND SAND, ELECTRIC-MOTOR-DRIVEN, TRAILER MOUNTED, 150 TONS PER HOUR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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